Session 7: Linear Models, Gradient Descent, the XOR Problem & Secondary Structure Prediction

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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In gradient descent, each step moves the weights …

A along the gradient: uphill

B against the gradient: downhill

C towards zero

D at random

Answer now
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Which logic function is true when exactly one of two inputs is true?

A AND

B OR

C XOR

D NAND

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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Which structure is held together by hydrogen bonds from residue i to residue i + 4?

A β-strand

B α-helix

C Coil

D Disulphide bond

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

A weighted sum	One sum or two	Biology and input
perceptron, gradient descent	AND, OR, XOR, hidden layer	GOR, PHD, composition

Answer now

Scan the code, or go to **quiz.bioinfo.se/live**

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What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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Two pictures behind every name today

One weighted sum: one straight boundary



PSSM score (no activation) · perceptron (step) · logistic regression (sigmoid) → AND, OR, NAND; not XOR

Add a hidden layer: several boundaries, combined



multilayer perceptron (MLP), hidden units with tanh or ReLU, e.g. PHD → XOR

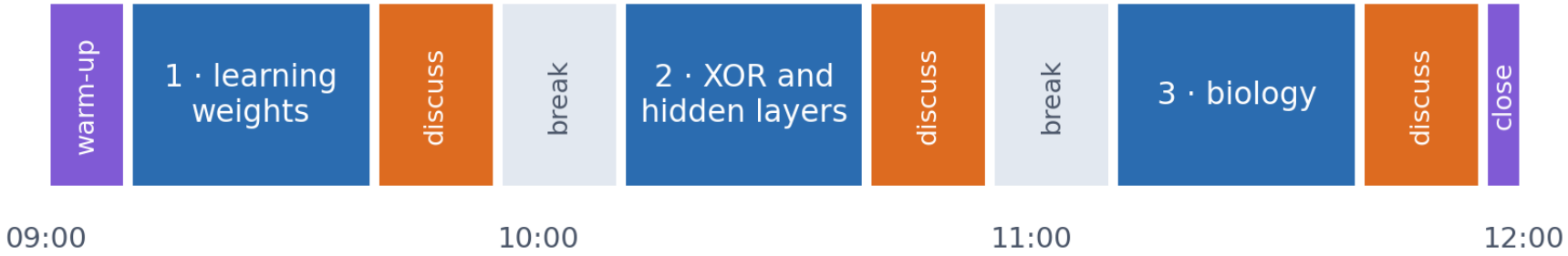
Without the nonlinear activation, the two weighted sums collapse into one, and XOR is impossible again.

PSSM, perceptron, linear and logistic regression, sigmoid, tanh, hidden layer, MLP, PHD: each is one of these two pictures.

See course.bioinfo.se/session07.html, 'Two pictures to keep in mind'.

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Three blocks: about 30 min talk and 15 min discussion



Block 1: a weighted sum, and how to learn its weights. Block 2: what one weighted sum can and cannot do (XOR), and the hidden layer. Block 3: secondary structure, and why the input matters as much as the model.

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

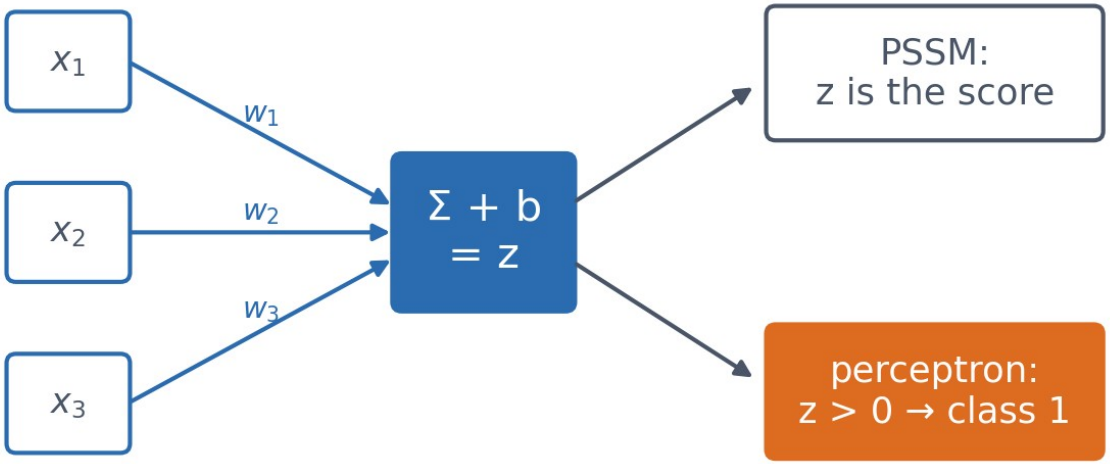
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BLOCK 1 · ABOUT 30 MIN

A weighted sum, learned from examples

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You already know the core of a neural network



- $z = w_1x_1 + \ldots + w_nx_n + b$
- The profile score of Session 5
- Bias b: moves the threshold

See course.bioinfo.se/session07.html, 'A profile score and a perceptron: the same weighted sum'.

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Rosenblatt 1958: a perceptron modelled on a neuron

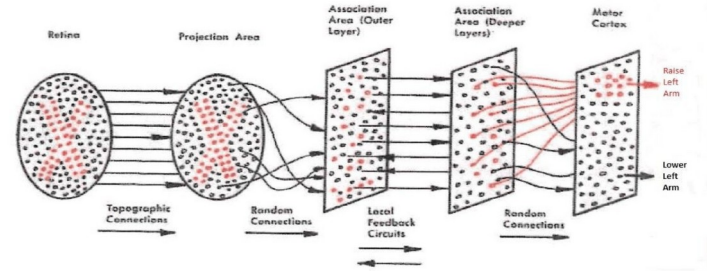


FIG. 1 — Organization of a biological brain. (Red areas indicate active cells, responding to the letter X.)

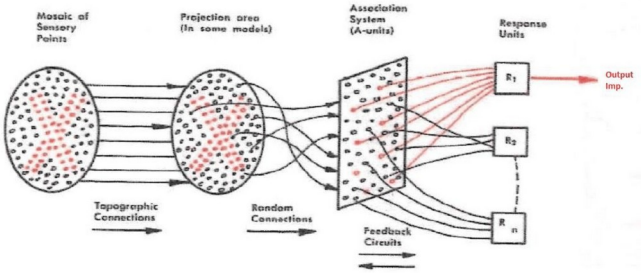
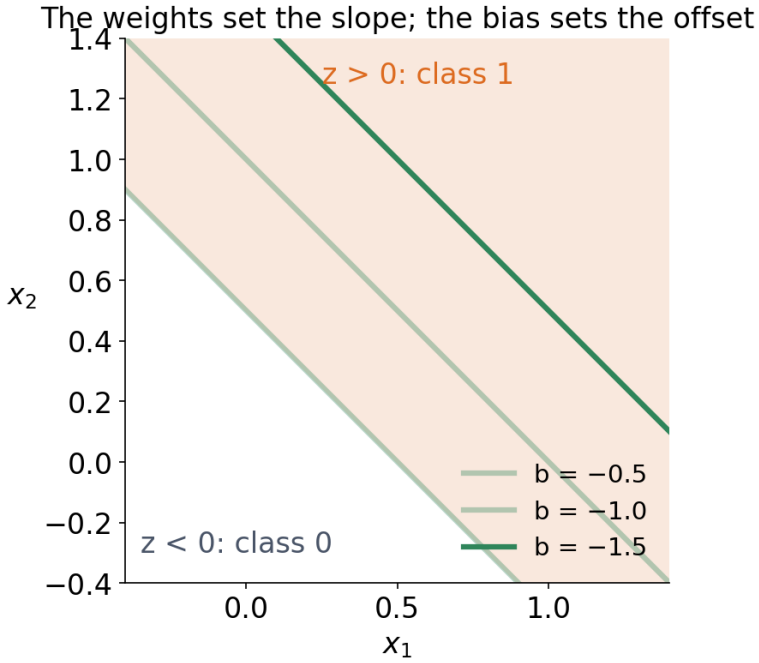


FIG. 2 — Organization of a perceptron.

Rosenblatt's own diagram: a brain circuit (top) and the perceptron built to mirror it (bottom). Weights = connection strengths; the cell fires when the weighted sum crosses a threshold.

Rosenblatt (1958), Research Reviews, Office of Naval Research; public domain, via Wikimedia Commons.

z = 0 is a line: the decision boundary



- The weights set the slope
- The bias slides the line
- In more dimensions: a plane

See course.bioinfo.se/session07.html, 'A profile score and a perceptron: the same weighted sum'.

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Two ways to get the weights

Count (Session 5)

PSSM

- Count residues in an alignment
- Turn counts into log-odds
- Needs a trusted alignment

Learn (from now on)

gradient descent

- Start with arbitrary weights
- Measure the loss on labelled examples
- Step every weight downhill

See course.bioinfo.se/session07.html, 'Learning the weights from examples'.

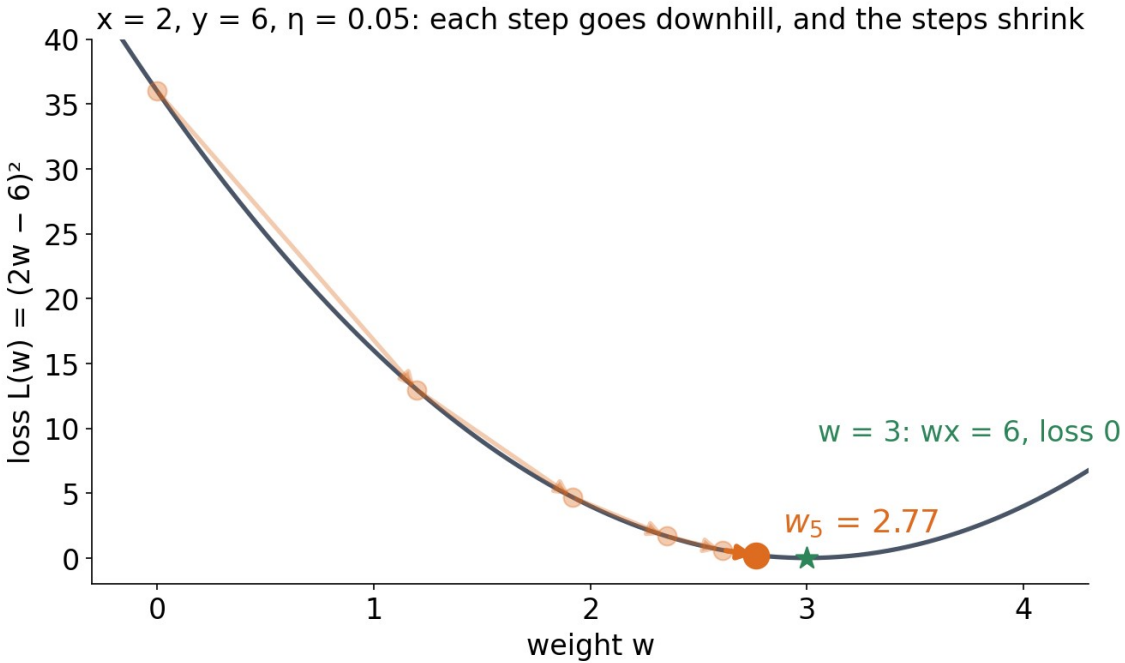
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What a loss function measures

- How wrong the model is, as one number
- **Squared error: $L(w) = (wx - y)^2$**
- Small loss = predictions close to the labels
- It must be differentiable — that is the whole requirement
- The step activation is not. Hence the sigmoid.

See course.bioinfo.se/session07.html, 'Learning the weights from examples'.

Gradient descent: one weight, one example



- **dL/dw = 2(wx − y)·x**
- Heads for w = 3
- **The steps shrink by themselves**

See course.bioinfo.se/session07.html, the worked example and its table.

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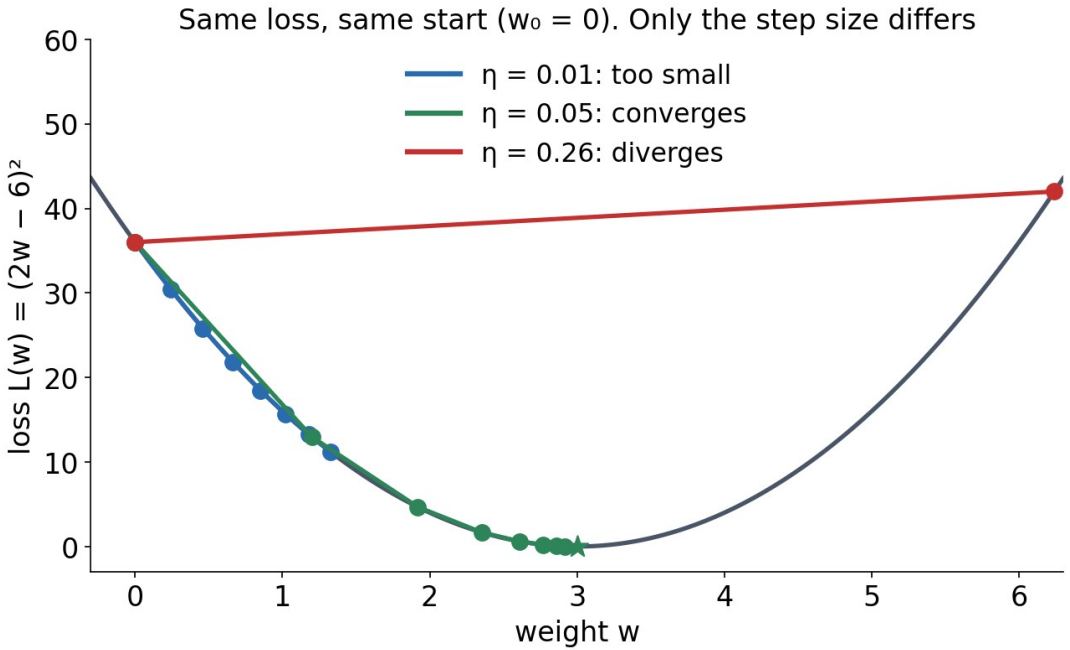
The same three steps, as arithmetic

- n = 0: w = 0.00, dL/dw = 2(0 − 6)(2) = −24.0 → 1.20
- n = 1: w = 1.20, dL/dw = 2(2.4 − 6)(2) = −14.4 → 1.92
- n = 2: w = 1.92, dL/dw = 2(3.84 − 6)(2) = −8.64 → 2.35
- **The gradient shrinks, so the steps shrink**
- η stays 0.05 throughout

Table 1 on course.bioinfo.se/session07.html.

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The learning rate decides whether you get there

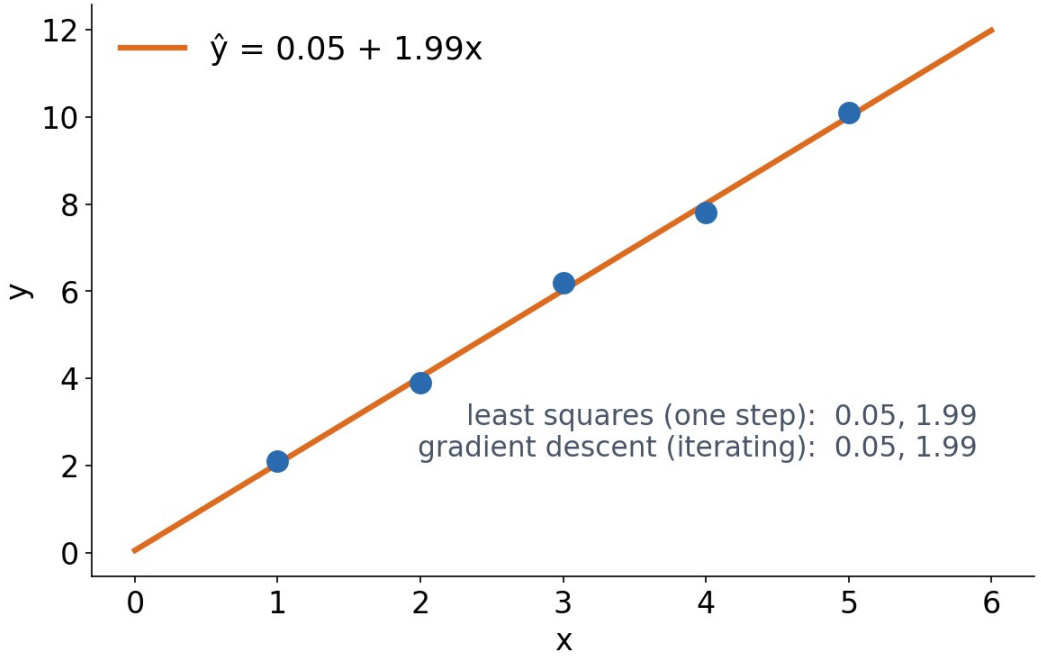


η = 0.26 multiplies the distance by −1.08. The sign flips, so it overshoots, and the size grows, so it overshoots further every time.

Same example as the previous slide; compare Session 1's second discussion problem.

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Derived or learned: the same line



- **Linear regression has a direct solution**
- Gradient descent reaches the same b_0, b_1
- Most models have no direct solution

See course.bioinfo.se/session07.html, 'Deriving vs. learning, on the simplest possible model'.

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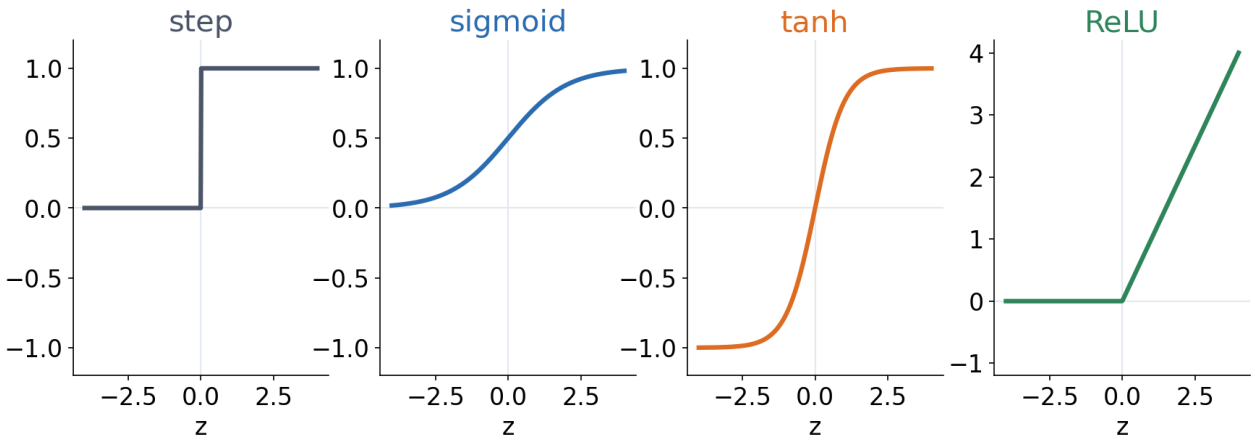
One example, or all of them?

- The worked example used a single point
- **Real training averages the loss over many examples**
- All of them at once: slow but steady
- A small random batch: noisier, far faster
- SGD, momentum, Adam — Session 9

See course.bioinfo.se/session07.html, 'Learning the weights from examples', and its further reading.

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Activation functions: what happens after the sum



- Step: 1 if z > 0, else 0 — a class (perceptron)
- Sigmoid: a probability (logistic regression)
- tanh, ReLU: inside hidden layers (Block 2)

See course.bioinfo.se/session07.html; logistic regression returns in Session 8.

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In the lab: raw tensors, on purpose

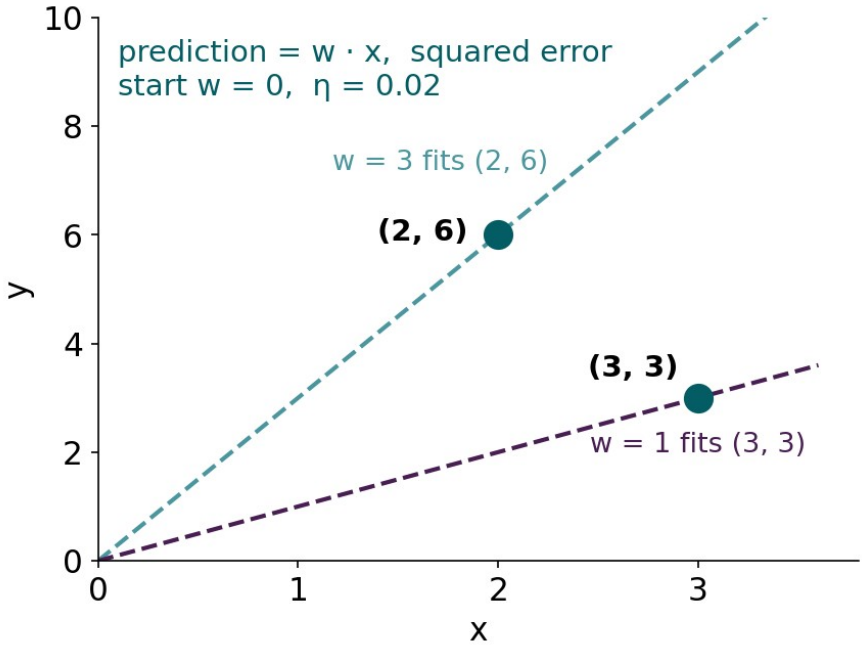
- Autograd from Session 2, nothing above it
- **No nn.Module, no training-loop helper**
- Every step of the mechanics stays visible
- Session 8 hands you the shortcuts

See course.bioinfo.se/session07.html, 'Learning the weights from examples'.

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Where does gradient descent take w?

Two examples, (2, 6) and (3, 3); prediction = w · x. Start at w = 0, η = 0.02.



A To w = 3: it fits the first example

B To w = 2: halfway between the two

C To about 1.6: closer to w = 1

D Nowhere: w keeps jumping

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Where does gradient descent take w?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Where does gradient descent take w?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-1.ipynb

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Where does gradient descent take w?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Where does gradient descent take w?

Results, and who explains

Both rounds side by side.

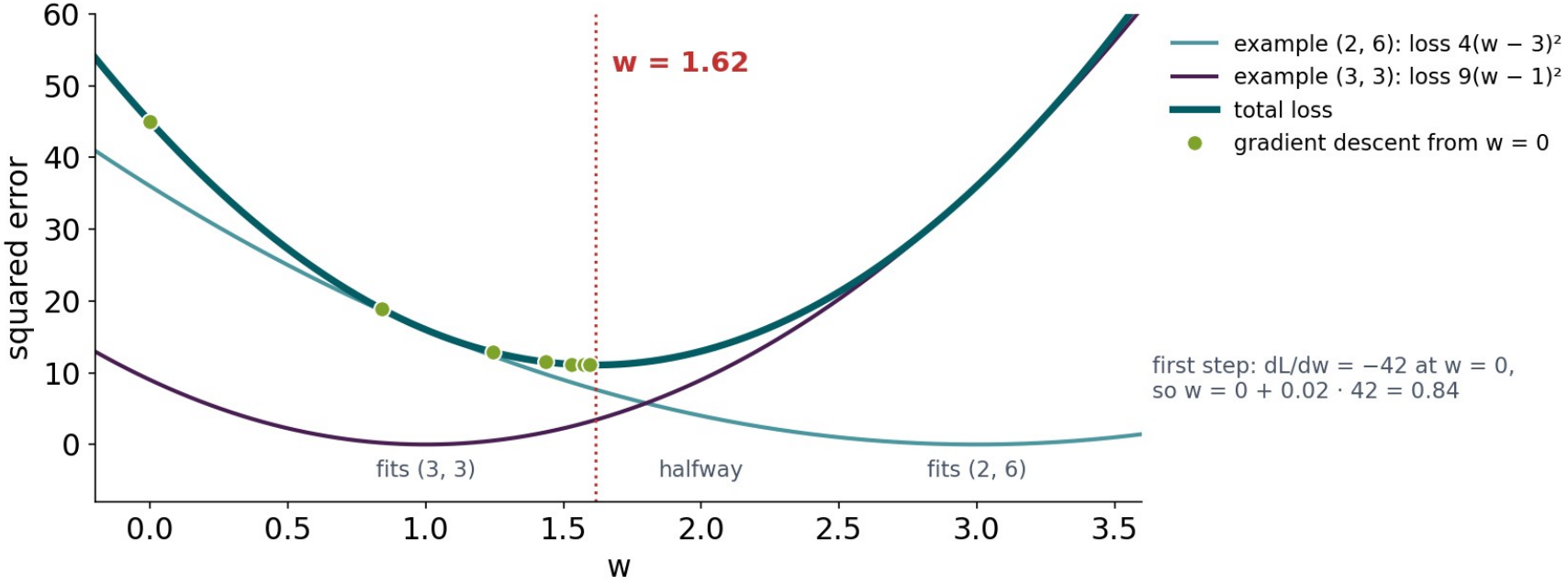
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-1.ipynb

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The answer: about 1.6, a compromise that leans to the steeper loss



Total loss $4(w-3)^2 + 9(w-1)^2$. The example (3, 3) has the larger x, so its loss is steeper and pulls harder: $w^* = (4{\cdot}3 + 9{\cdot}1)/13 = 1.62$, not the plain average 2. No weight fits both examples, so gradient descent finds the best compromise.

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BLOCK 1 DONE

Break

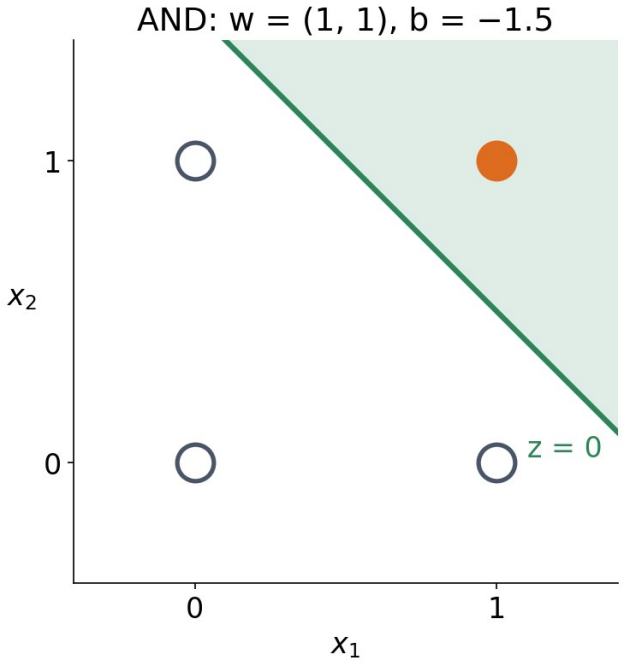
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BLOCK 2 · ABOUT 30 MIN

What one weighted sum can and cannot do

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AND is one line through the four corners



- **Output 1 if z > 0, else 0**
- w = (1, 1), b = −1.5
- Only (1, 1) lies on the green side

See course.bioinfo.se/session07.html, 'What one weighted sum can represent'.

SciLifeLab

Find weights for a perceptron that implements OR

- Sign activation: output +1 if z > 0, else −1
- **1. Find w_1, w_2, b so z reproduces the OR truth table**
- **2. Hint: where must the line sit relative to the four corners?**
- **3. Is your solution the only one that works?**

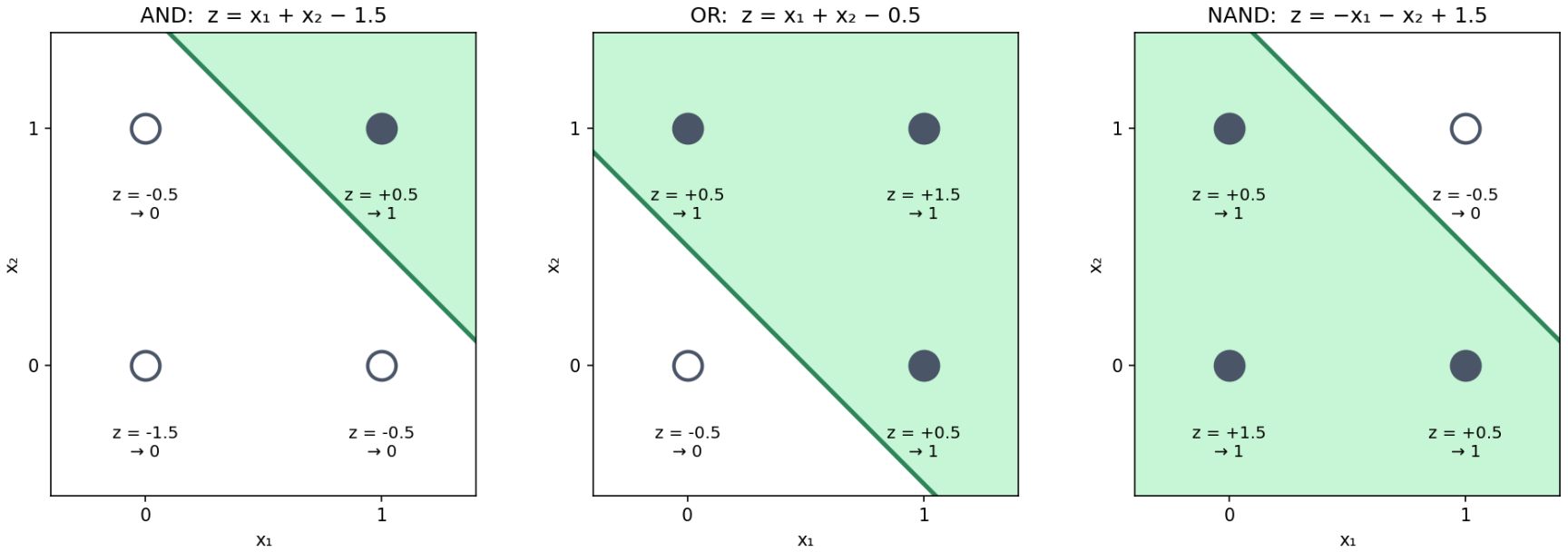
Scan me

Opens in Colab

Discussion notebook 2: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-2.ipynb

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AND, OR, NAND: each is one line



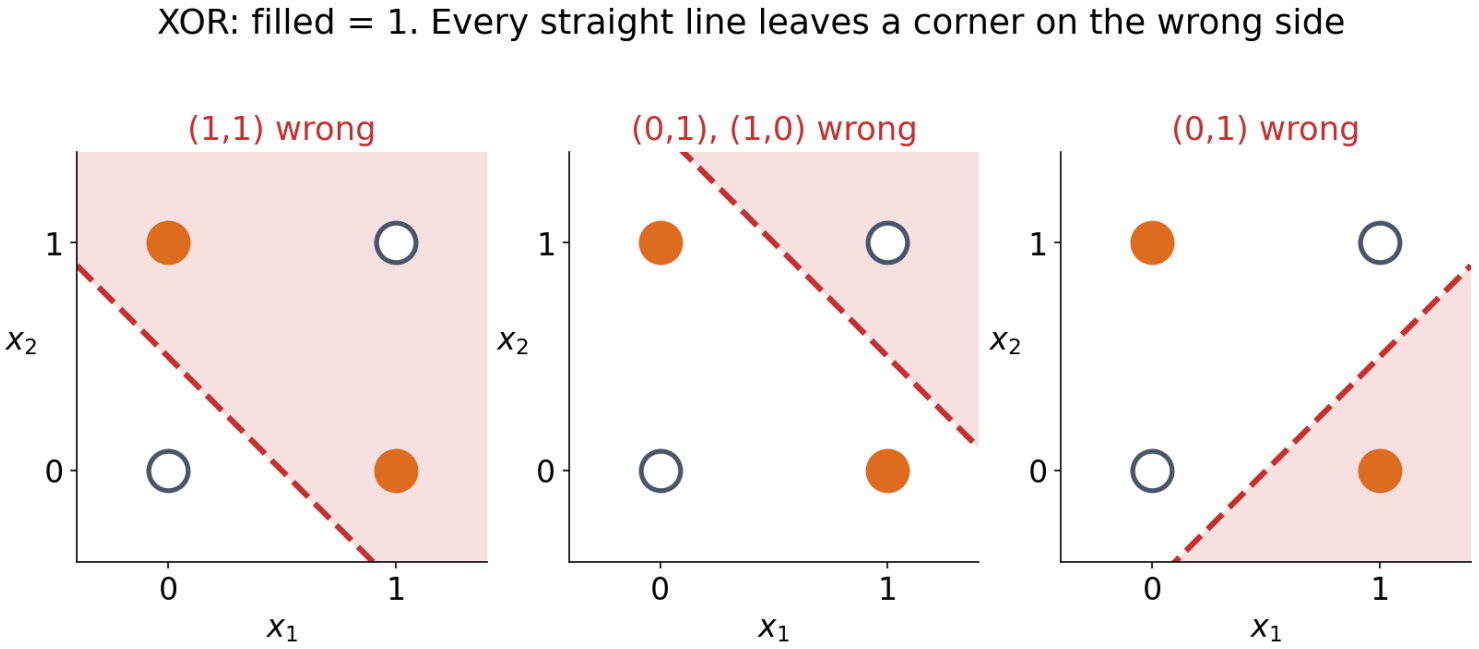
Filled circle: output 1. Open circle: output 0. Green line: z = 0, the decision boundary; the shaded side has z > 0.

Filled: output 1, open: output 0, z under each corner. Green line: z = 0; shaded side: z > 0.

See course.bioinfo.se/session07.html, Section 7.5, and notebook Section 3.

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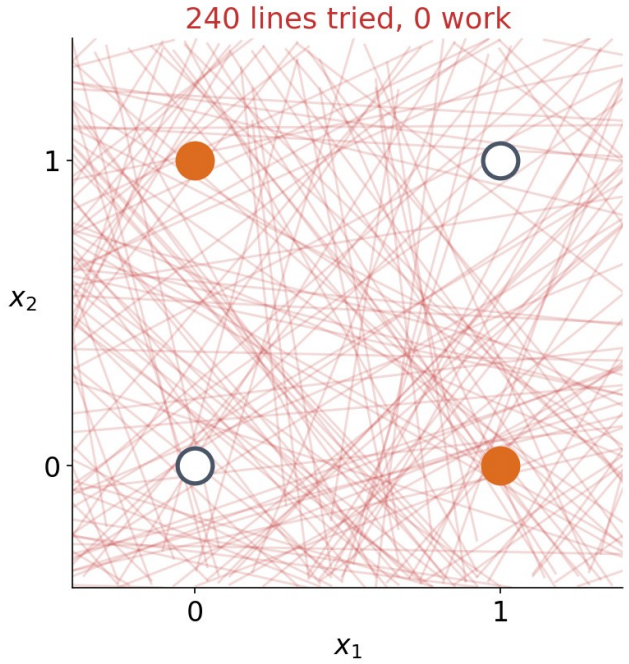
XOR: try to draw the line



XOR outputs 1 when the inputs differ. Its two 1-corners sit diagonally opposite, and no straight line separates one diagonal from the other.

See course.bioinfo.se/session07.html, 'What one weighted sum can never represent: XOR' (the algebraic proof is optional there).

The notebook tries 226,981 perceptrons

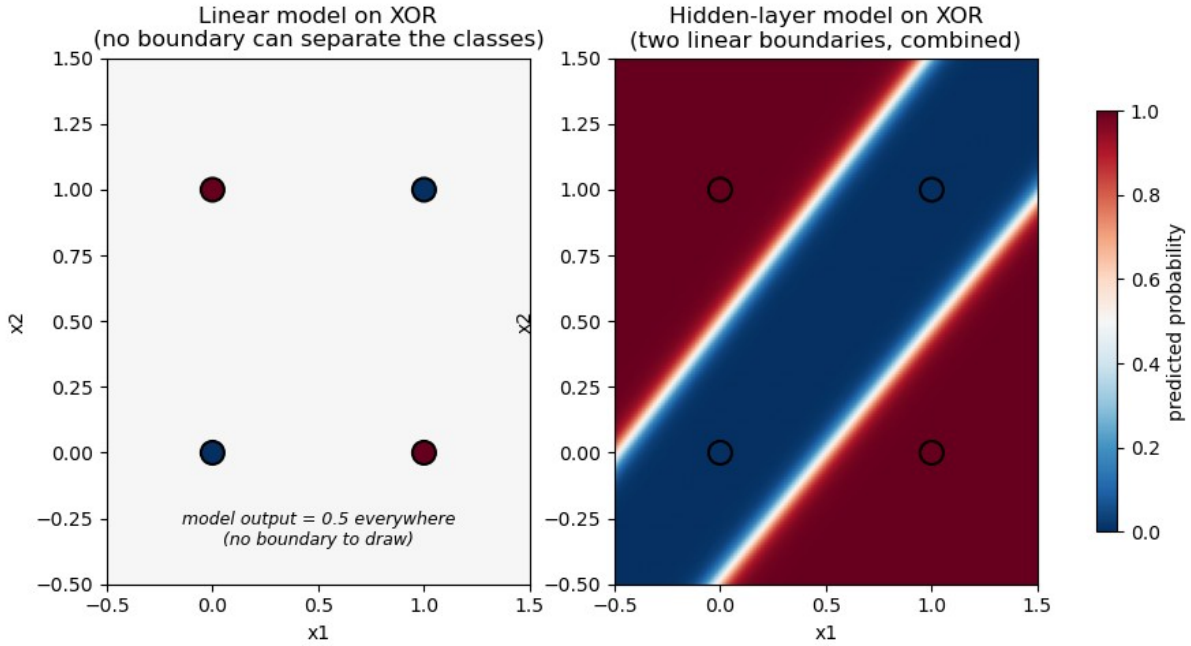


- The notebook tries 226,981
- **Not one of them works**

Notebook Section 4.

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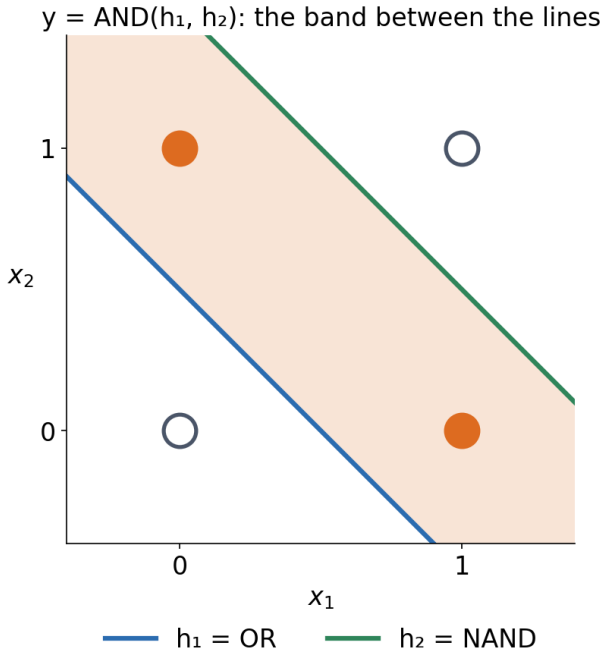
Gradient descent gives up: 0.5 everywhere



A linear model trained on XOR settles on predicting 0.5 for all four inputs (loss ln 2): any line it draws gets at least one corner wrong.

Notebook Section 4; course.bioinfo.se/session07.html.

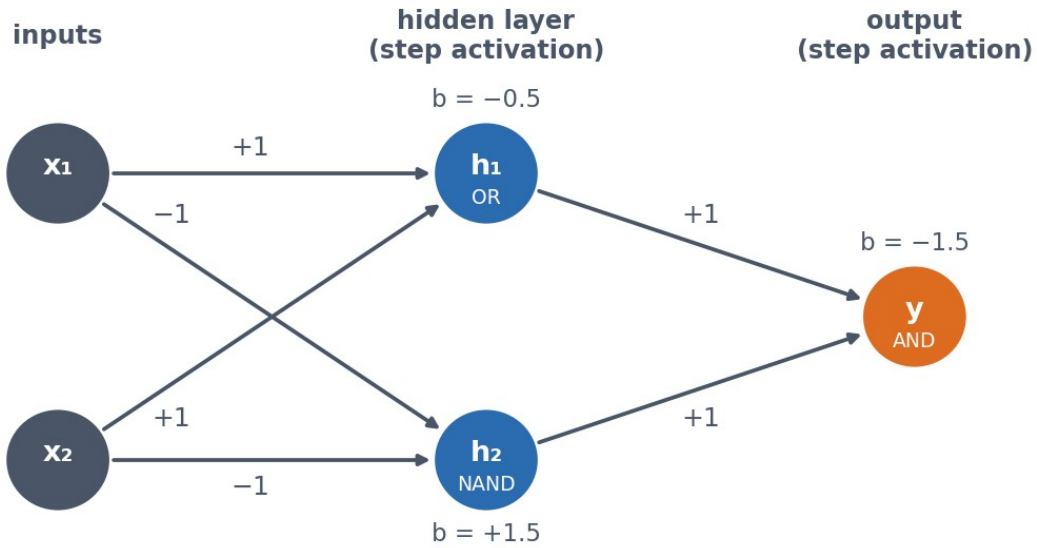
Two lines make XOR



- **Hidden unit 1: OR**
- **Hidden unit 2: NAND**
- Output: AND of the two
- 1 only in the band between the lines

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A two-layer perceptron for XOR



each unit: z = weighted sum of its inputs + b, output = 1 if z > 0, else 0

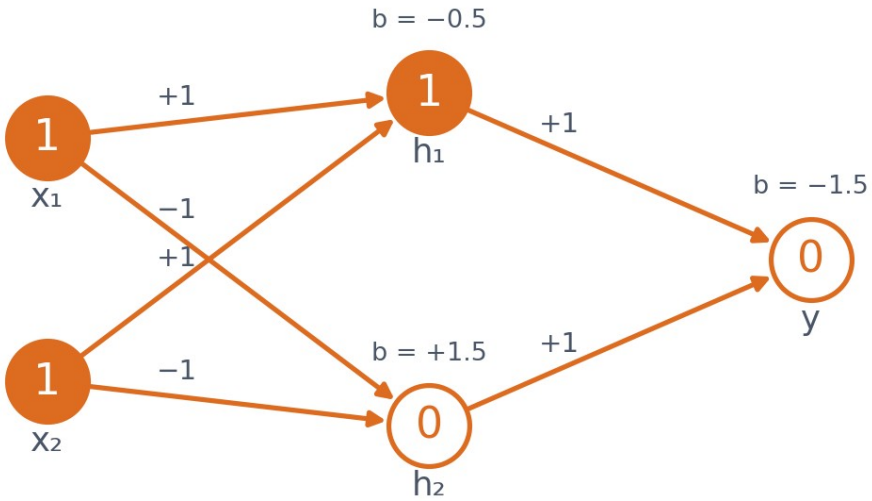
h_1 = OR (+1, +1, bias −0.5), h_2 = NAND (−1, −1, +1.5), y = AND(h_1, h_2) (+1, +1, −1.5). Every unit: weighted sum + bias, then step.

See course.bioinfo.se/session07.html, Section 7.7.

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Step by step through the network

y = AND(h₁, h₂): only one input is on, so y = 0 — which is XOR(1,1)

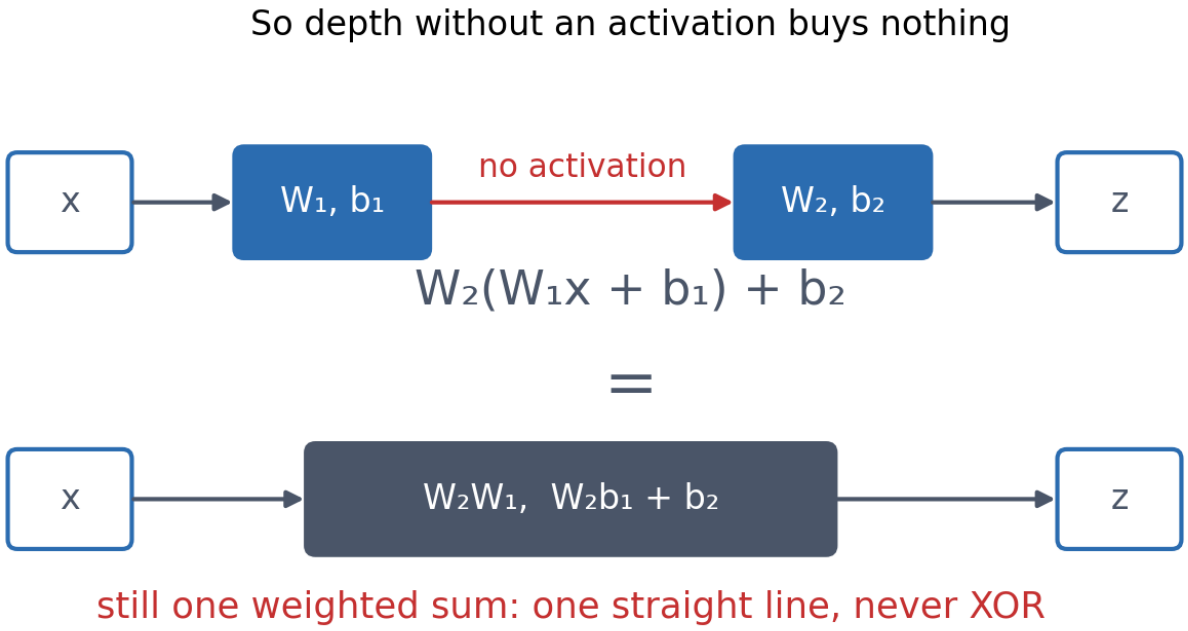


The output unit is AND: it needs both. Only h_1 is on, so y = 0 — and XOR(1, 1) is indeed 0.

Notebook Section 5 computes this in PyTorch and prints every value.

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Without an activation, two layers are one



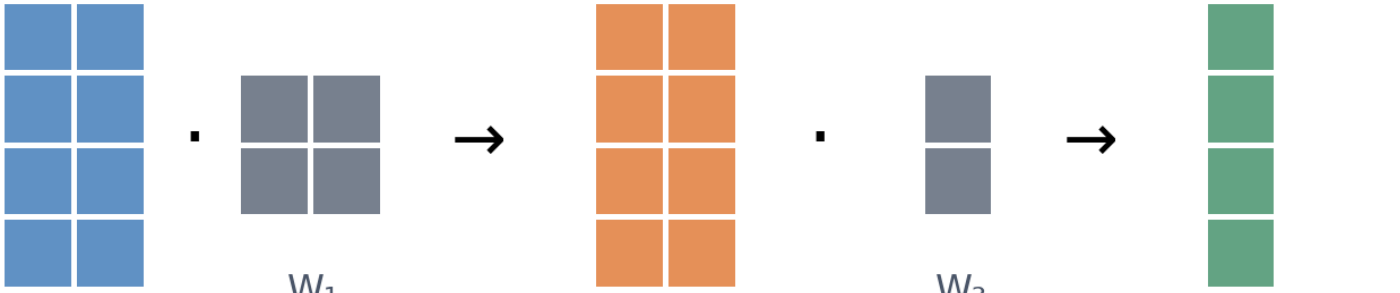
So two layers without an activation are one layer — one straight boundary, and never XOR. The activation is what prevents this.

See course.bioinfo.se/session07.html, 'The activation must be nonlinear'.

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Shapes in the notebook's XOR network

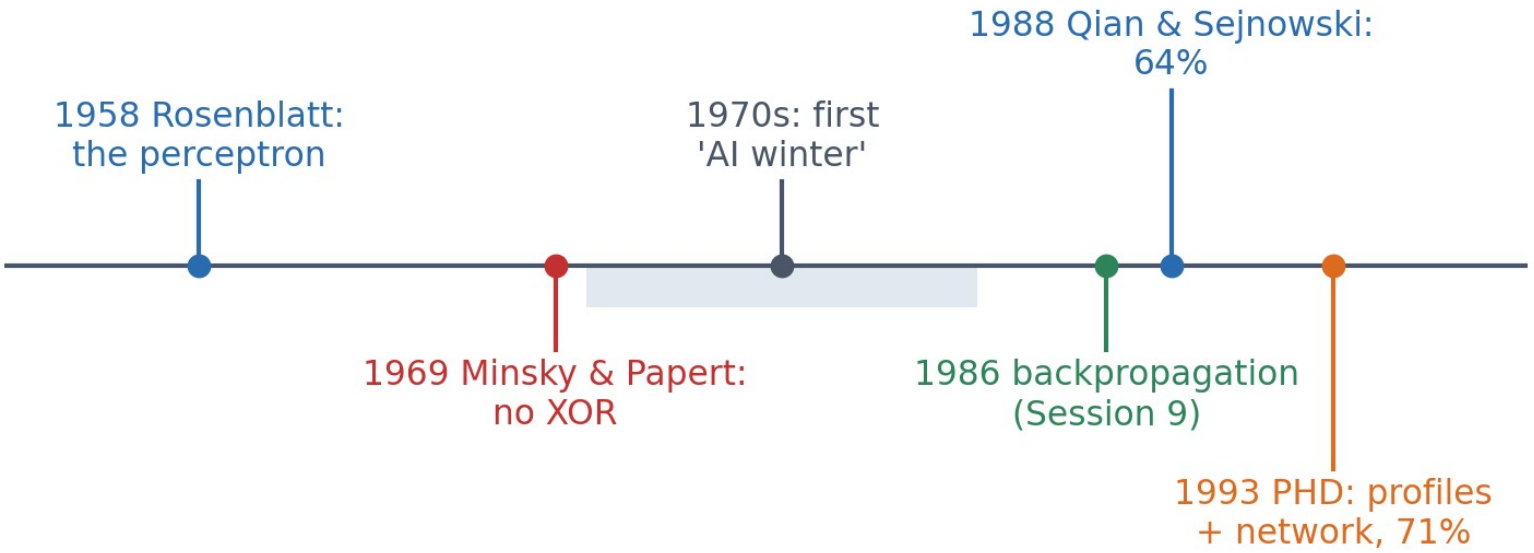
4 examples · 2 inputs → 2 hidden values each → 1 prediction each



Follow the shapes (Session 1's rules) and the training loop is this picture in code.

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A short history



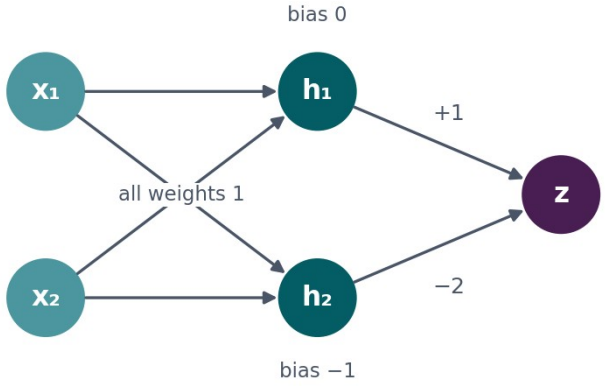
Minsky & Papert proved that single-layer perceptrons cannot solve XOR-like problems. Training through hidden layers became practical with backpropagation (Session 9).

See course.bioinfo.se/session07.html, 'A short history'.

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Which logic function does the network compute?

$h_1 = \mathrm{ReLU}(x_1 + x_2)$, $h_2 = \mathrm{ReLU}(x_1 + x_2 - 1)$, $z = h_1 - 2h_2$, for x_1, x_2 in $\{0, 1\}$.



$$h_1 = \mathrm{ReLU}(x_1 + x_2)$$

$$h_2 = \mathrm{ReLU}(x_1 + x_2 - 1)$$

$$z = h_1 - 2\,h_2$$

$\mathrm{ReLU}(t) = \max(0, t)$

A AND

B OR

C XOR

D None: untrained weights compute nothing useful

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Which logic function does the network compute?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Which logic function does the network compute?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-3.ipynb

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Which logic function does the network compute?

Answer again

Same question, after the discussion.

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Which logic function does the network compute?

Results, and who explains

Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 3 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-3.ipynb

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The answer: XOR, and only with the ReLU

x_1 x_2	with ReLU	without ReLU	XOR
0 0	0	2	0
0 1	1	1	1
1 0	1	1	1
1 1	0	0	0
	= XOR	not XOR	

Without ReLU: $z = (x_1 + x_2) - 2\,(x_1 + x_2 - 1) = 2 - x_1 - x_2$

For any weights: $W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$

Layers without an activation multiply out to one weighted sum: one line, never XOR.

With ReLU the network gives 0, 1, 1, 0: XOR. The unfixed notebook had no activation and gave 2, 1, 1, 0. Without an activation, layers multiply out to one weighted sum for any weights, and one weighted sum draws one line: never XOR.

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BLOCK 2 DONE

Break

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BLOCK 3 · ABOUT 30 MIN

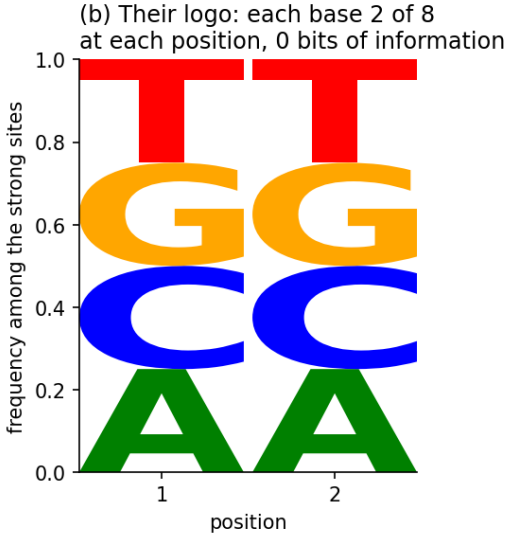
Biology, and why representation matters

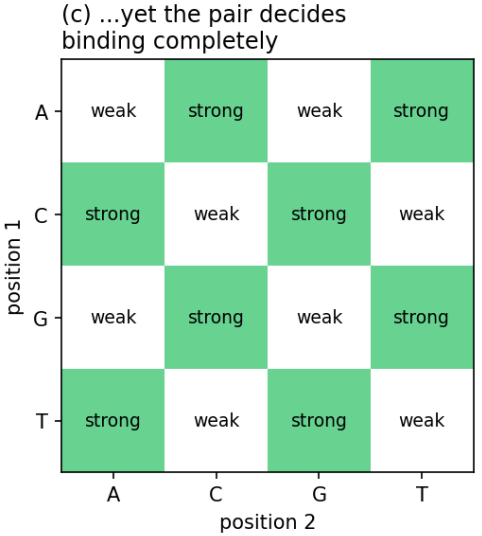
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Bulyk et al. (2002): binding positions are not independent

(a) The data: the 16 possible sites

strong (R Y or Y R)			weak (R R or Y Y)		
A	C	RY	A	A	RR
A	T	RY	A	G	RR
C	A	YR	C	C	YY
C	G	YR	C	T	YY
G	C	RY	G	A	RR
G	T	RY	G	G	RR
T	A	YR	T	C	YY
T	G	YR	T	T	YY





A toy motif: strong binding with one purine and one pyrimidine. (a) The 16 sites. (b) The logo of the strong sites: each base 2 of 8, 0 bits... (c) ...yet the pair decides binding completely. A logo, like a profile, sees one position at a time.

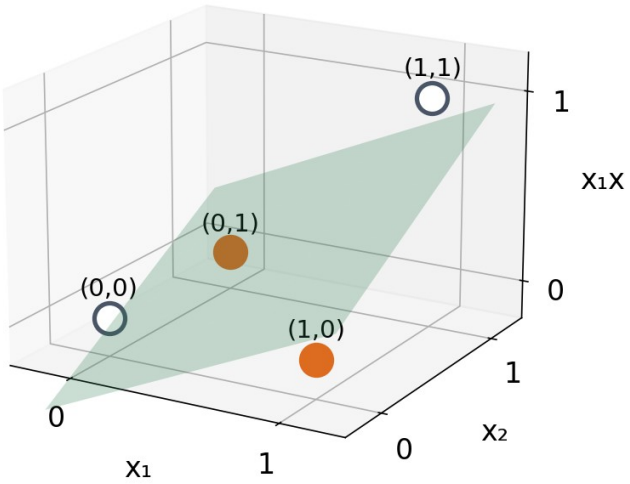
Bulyk, Johnson & Church (2002), Nucleic Acids Res 30:1255. The toy motif is illustrative, not the paper's data.

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Another fix: change the input, not the model

With the input x_1x_2, one flat plane separates XOR

$$z = x_1 + x_2 - 2x_1x_2$$



- **Add x_1x_2 as a third input**
- XOR = $x_1 + x_2 - 2x_1x_2$: linear
- But someone must know x_1x_2 matters
- A hidden layer learns such features

See course.bioinfo.se/session07.html, 'A different fix: change the input'.

Helix, strand, coil: a label for every residue

Ubiquitin (PDB 1UBQ)

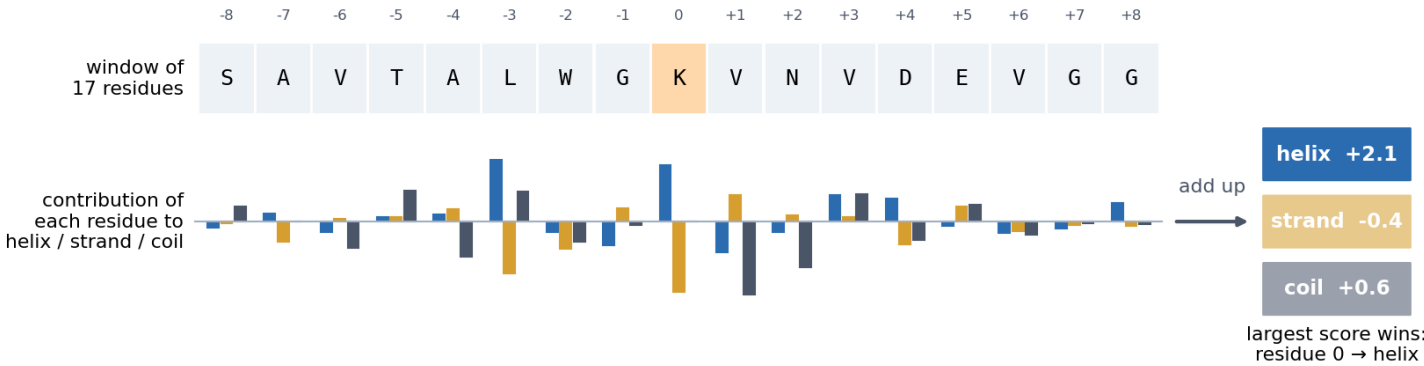


helix strand coil

- **Helix: H-bonds 4 residues apart**
- **Strand: H-bonded edge to edge in a sheet**
- Coil: everything else
- Predict one of three labels per residue

See course.bioinfo.se/session07.html, 'Secondary structure prediction: PHD'.

GOR (1978): three profiles over a 17-residue window



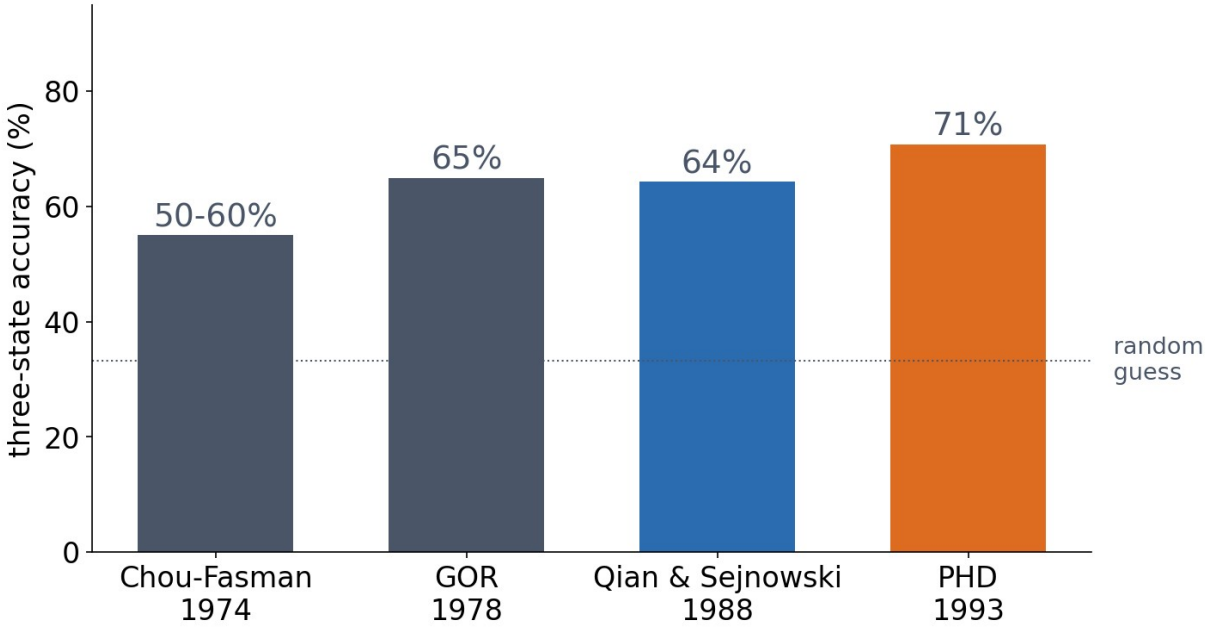
score(state) = Σ over the 17 positions of a table value for (state, offset, amino acid): three position-specific scoring matrices, one per state — a linear model.

Bar heights are illustrative; GOR estimated its tables from the known structures of the time.

Each residue in the window adds a value to the helix, strand and coil scores; the largest total wins. Three weighted sums: a linear model, picture 1.

Garnier, Osguthorpe & Robson (1978), J Mol Biol 120:97. Background for the capstone project (optional box on the page).

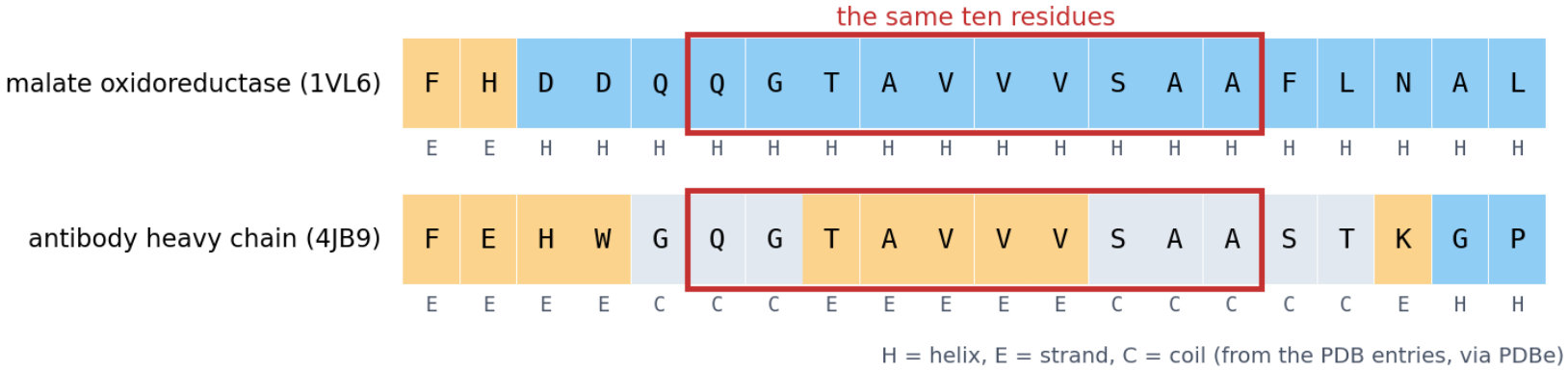
Single sequences got stuck at about 65%



- Rules, windows, even a neural network
- **Qian & Sejnowski 1988: 64%**
- Then PHD: 71%. What changed?

Historical consensus figures (Wikipedia, 'Protein structure prediction'); the papers used different test sets.

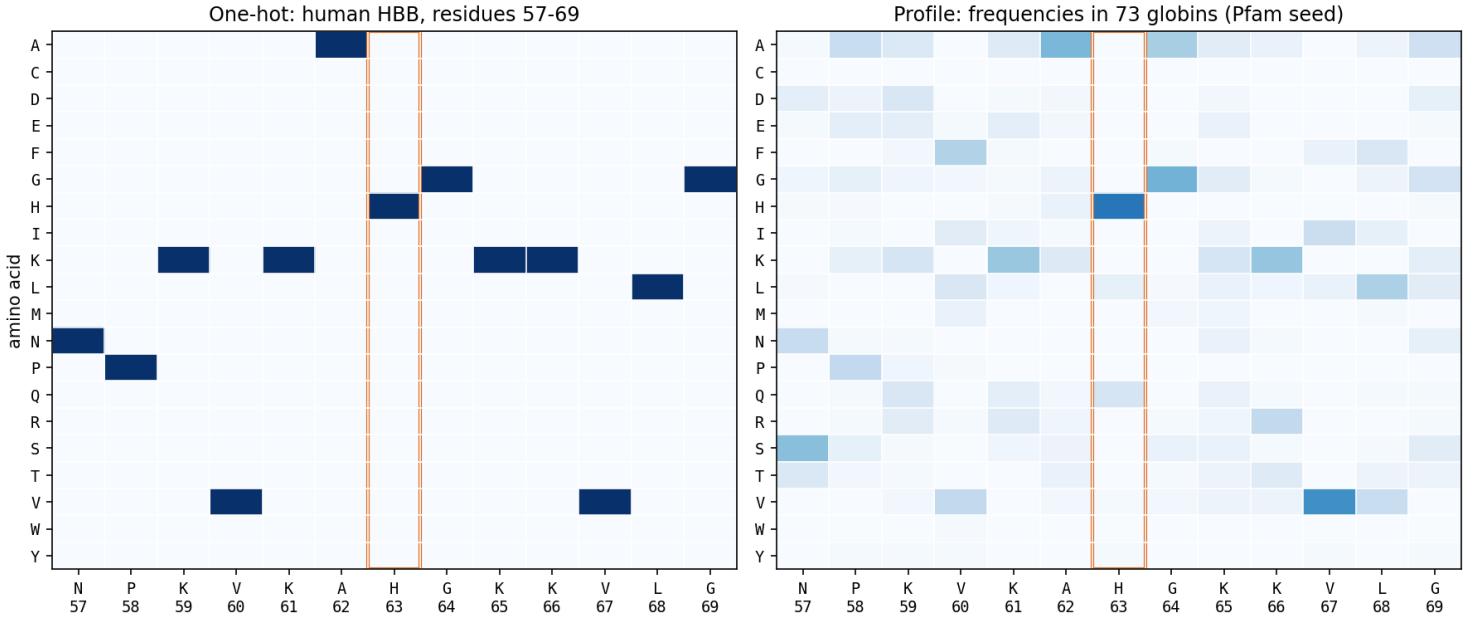
Same ten residues, different structure



QGTAVVVSAA: helix in a malate oxidoreductase (1VL6), strand and coil in an antibody heavy chain (4JB9). A window of local sequence alone cannot decide.

Li et al. (2015), ChSeq, Protein Sci 24:1075; Kabsch & Sander (1984), PNAS 81:1075.

From letters to numbers: one-hot, and profiles



Each column is one position: 20 numbers. One-hot puts a single 1 at the residue that is there; the profile spreads the column over the residues its homologs have.

One residue = 20 numbers with a single 1 (not A = 1, C = 2, which implies a false order). A profile column is the average of the aligned sequences' one-hot vectors.

Profile from the Pfam globin seed alignment (Session 4).

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PHD (1993): the input was the innovation

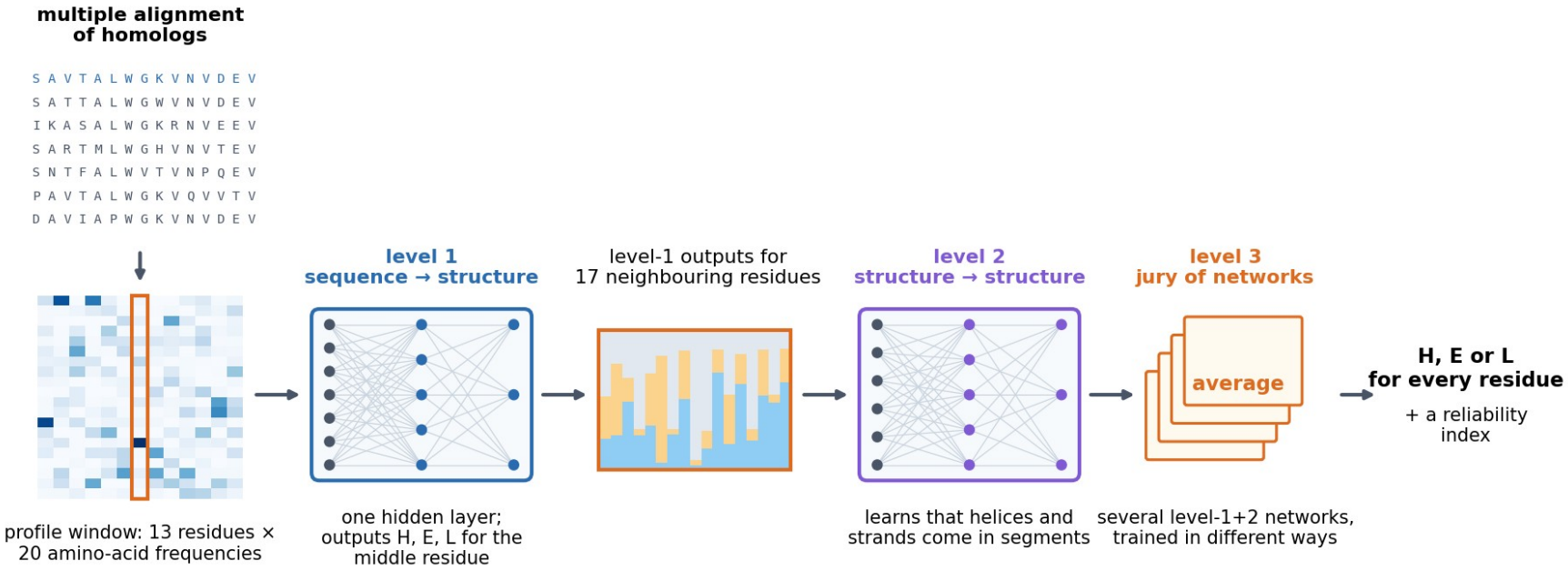
- **A profile from a multiple alignment of homologs, not a single sequence**
- Structure is more conserved than sequence
- The family shows what one sequence hides

+6-8
points from the alignment alone

70.8%
three-state accuracy

Rost & Sander (1993), J Mol Biol 232:584-599; Illergård, Ardell & Elofsson (2009).

PHD: three levels

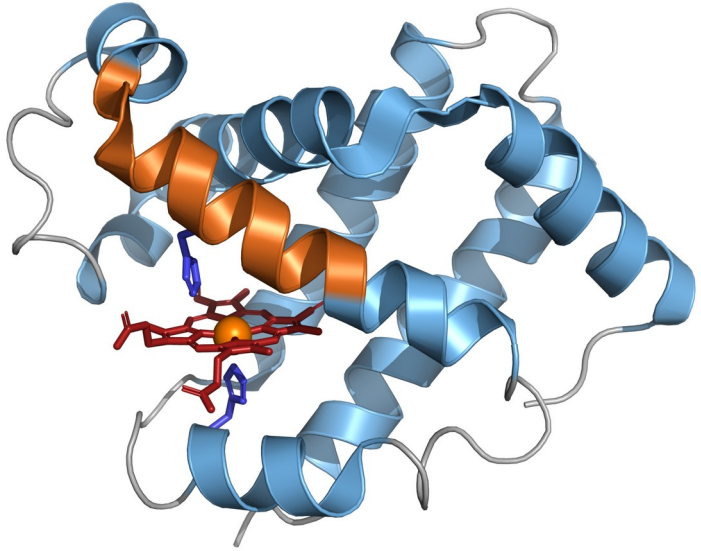


Level 1: profile window of 13 residues → network → H/E/L. Level 2: level-1 outputs for 17 residues → network (segments of plausible length). Level 3: a jury averages several networks.

Rost & Sander (1993); Session 9 returns to PHD.

HBB: every residue has a label

- **2HHB chain B: 117 of 146 residues in helices**
- No strands
- Haem between the helices; His63 above, His92 below
- Orange: the 13-residue window around His63



HBB (2HHB, chain B)

helix blue, coil grey

Rendered with PyMOL; secondary structure from the PDB entry.

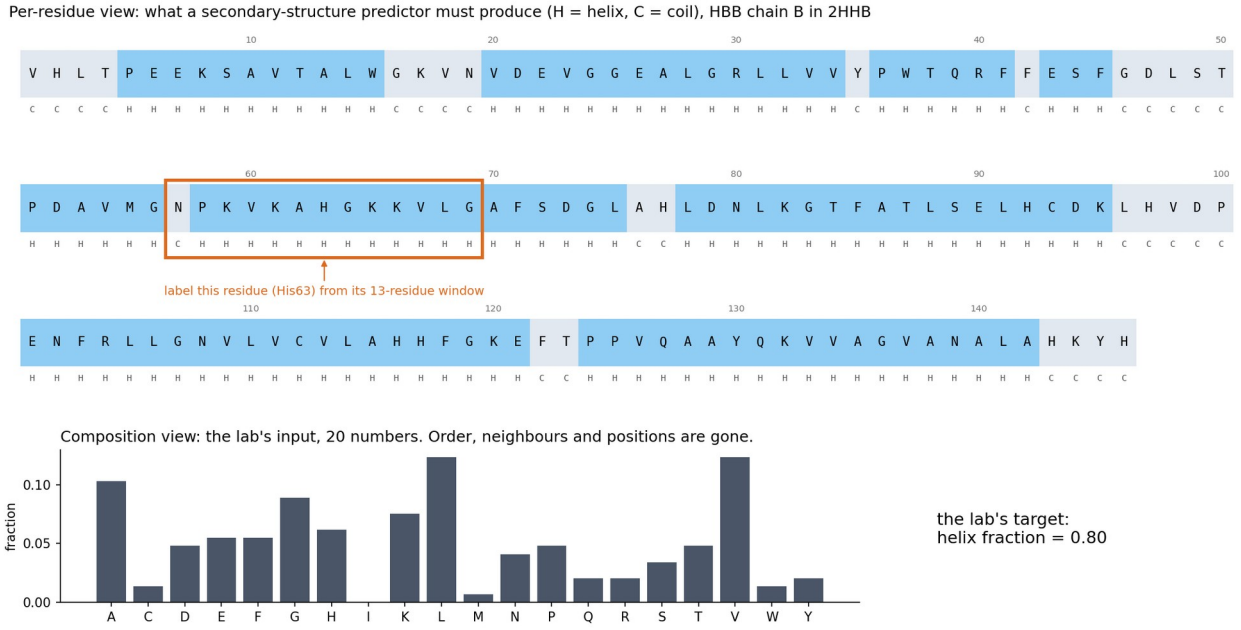
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Signal peptides: a weight matrix, then a network

- **1986, von Heijne: a weight matrix, −13 to +2**
- One weighted sum per window: picture 1
- **1997, SignalP 1.0: networks on sliding windows**
- Hidden layer: can combine positions, picture 2
- Early networks on a few dozen sequences overfitted (Session 8)

von Heijne (1986), NAR 14:4683; Nielsen et al. (1997), Protein Eng 10:1; history: Nielsen, Tsirigos, Brunak & von Heijne (2019), Protein J 38:200.

Two views of the same protein

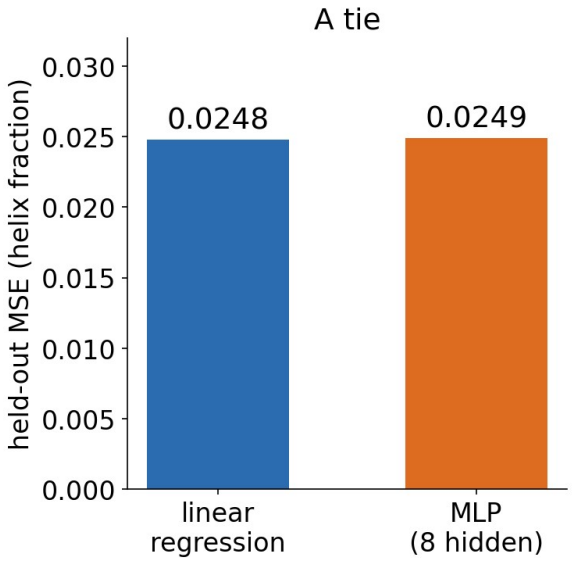


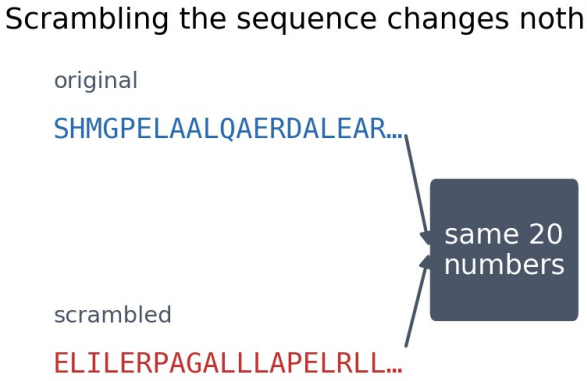
Top: per-residue labels and the window PHD reads to label His63. Bottom: the lab's input, 20 amino-acid fractions, and its single target, the helix fraction 0.80.

Secondary structure from EBI's PDBe API; tools/builders/make_session07_hbb_views.py.

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Representation matters as much as the model



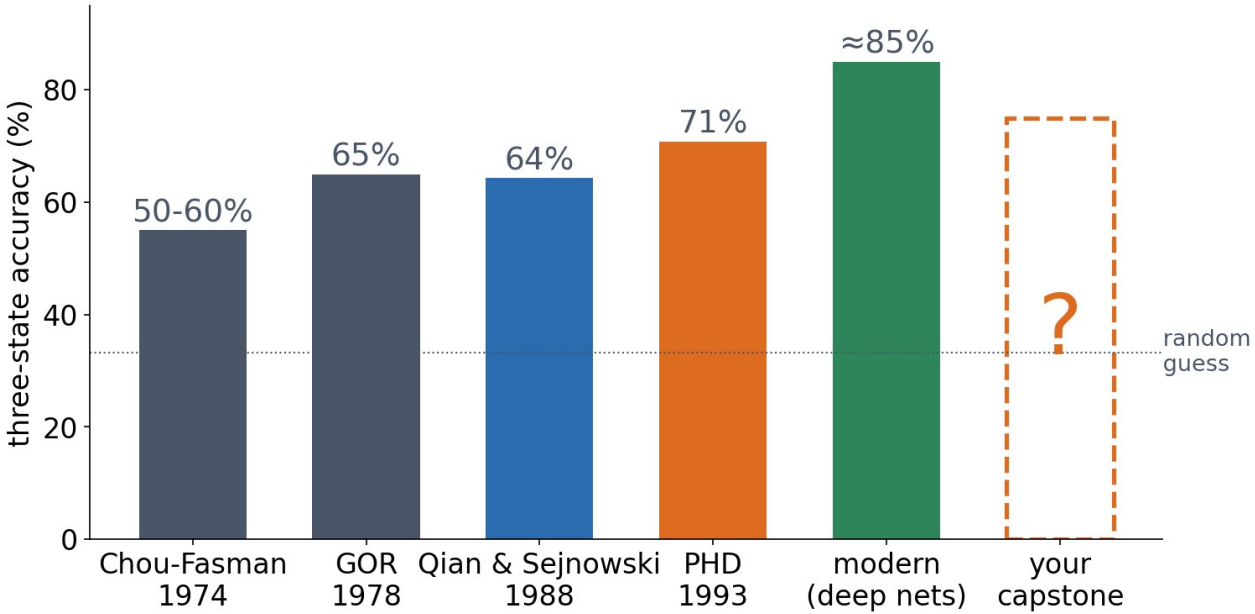


- **Linear and MLP: a tie**
- Composition has thrown away order
- No model recovers what is not in its input

Lab: 55 proteins from PDBe, 41 for training, 14 held out.

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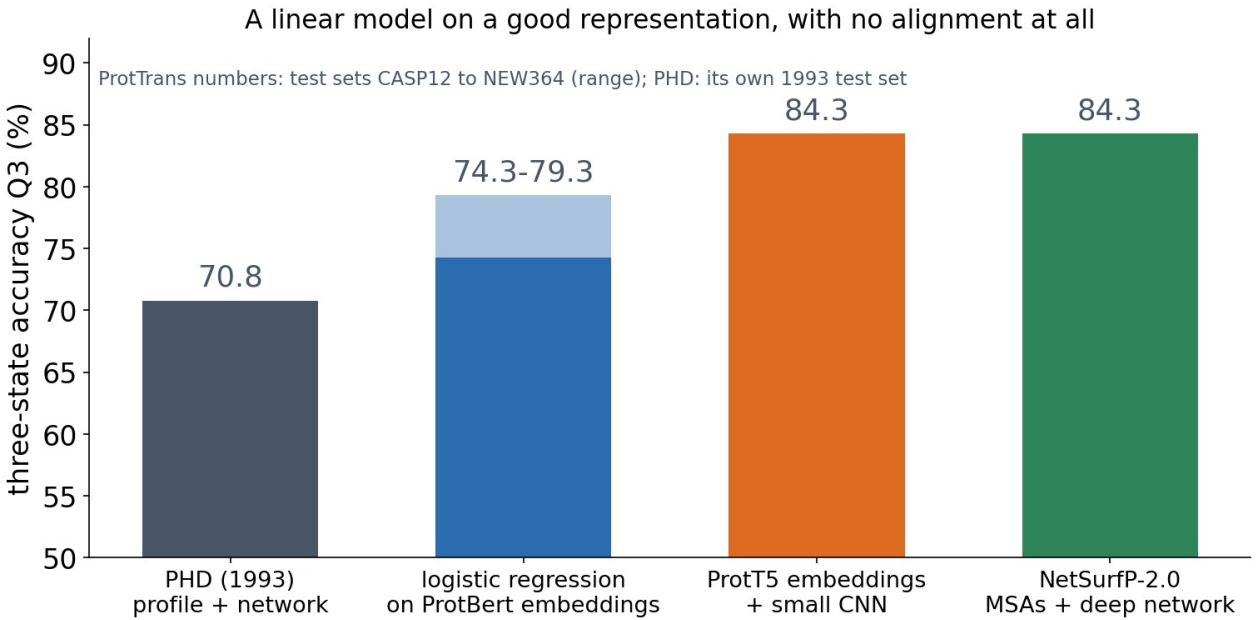
Can you do better?



- **Build your own predictor**
- Which input: sequence, profile, embedding?
- Which model?
- Tested on proteins it has never seen?

See the Capstone project page. https://course.bioinfo.se/capstone.html

Representation beats model, again



- Language-model embeddings per residue
- A linear model on them: 74-79%
- With a small CNN: as good as MSAs
- **If the input is good enough, how much model do you need?**

Elnaggar et al. (2022), IEEE TPAMI 44:7112 (ProtTrans); Rost & Sander (1993). Try it in the capstone project.

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What changed between the two pictures, and why does it allow XOR?

One weighted sum: one straight boundary



PSSM score (no activation) · perceptron (step) · logistic regression (sigmoid) → AND, OR, NAND; not XOR

Add a hidden layer: several boundaries, combined



multilayer perceptron (MLP), hidden units with tanh or ReLU, e.g. PHD → XOR

Without the nonlinear activation, the two weighted sums collapse into one, and XOR is impossible again.

Top: one weighted sum draws one straight boundary (AND, OR, NAND; never XOR). Bottom: hidden units with a nonlinear activation draw several boundaries, and the output combines them. Either one can only use what is in its input.

See course.bioinfo.se/session07.html, 'Summary: the two pictures again'.

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Which methods are picture 2?

Picture 2: a hidden layer with a nonlinear activation. Tick all that apply.

Picture 1



Picture 2



PSSM · perceptron · logistic regression · GOR · MLP · PHD

Which of these are picture 2?

A PSSM

B Perceptron

C Logistic regression

D GOR

E MLP

F PHD

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Which methods are picture 2?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Which methods are picture 2?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-4.ipynb

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Which methods are picture 2?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Which methods are picture 2?

Results, and who explains

Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 4 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session07-discussion-4.ipynb

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The answer: MLP and PHD; the others are one weighted sum

Picture 1: one weighted sum

PSSM: a score per residue, summed

perceptron: sum, then a step

logistic regression: sum, then a sigmoid

GOR: table values in a window, summed

Picture 2: a hidden layer

MLP: hidden units with a nonlinear activation

PHD: an MLP on profile windows,

plus a second network

An activation at the output (step, sigmoid) does not make a hidden layer. What matters is a nonlinear activation between two weighted sums.

An activation at the output (the perceptron's step, logistic regression's sigmoid) does not make a hidden layer. What makes picture 2 is a nonlinear activation between two weighted sums. Either picture can only use what is in its input: the lab's question.

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In up to three words: what was the hardest part of today?

A weighted sum	One sum or two	Biology and input
perceptron, gradient descent	AND, OR, XOR, hidden layer	GOR, PHD, composition

Answer now

Scan the code, or go to **quiz.bioinfo.se/live**



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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

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Session 8 next

From one yes/no output to multi-class: softmax regression
How do we know whether a model has learned anything useful?